

Image processing: Frequency domain

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This document covers some concepts related to frequency domain.

1 Proof of the Sifting Property of the Dirac Delta Function

The **sifting property** of the Dirac delta function, $\delta(t - t_0)$, states that the integral of the product of a function $f(t)$ and a delta function $\delta(t - t_0)$ "sifts out" the value of the function at the point where the delta function is non-zero. Mathematically, this is expressed as:

$$\int_{-\infty}^{\infty} f(t)\delta(t - t_0)dt = f(t_0)$$

1.1 Proof

The proof relies on the two fundamental defining properties of the Dirac delta function:

1. **It is zero everywhere except at a single point:** $\delta(t - t_0) = 0$ for $t \neq t_0$.
2. **Its total area is unity:** $\int_{-\infty}^{\infty} \delta(t - t_0)dt = 1$.

Let's begin with the integral we want to evaluate:

$$I = \int_{-\infty}^{\infty} f(t)\delta(t - t_0)dt$$

Because of the first property, the delta function $\delta(t - t_0)$ is only non-zero at $t = t_0$. This means the entire product $f(t)\delta(t - t_0)$ is zero for all t except at t_0 . Therefore, we can reduce the limits of integration to an infinitesimally small interval around t_0 . Let's call this interval $[t_0 - \epsilon, t_0 + \epsilon]$, where ϵ is an arbitrarily small positive number. The integral becomes:

$$I = \int_{t_0 - \epsilon}^{t_0 + \epsilon} f(t)\delta(t - t_0)dt$$

Within this very small interval, as long as the function $f(t)$ is continuous at t_0 , its value is approximately constant. We can approximate $f(t)$ within this tiny region by its value at the single point t_0 , i.e., $f(t) \approx f(t_0)$.

We can now factor out this constant value from the integral:

$$I = f(t_0) \int_{t_0-\epsilon}^{t_0+\epsilon} \delta(t - t_0) dt$$

The remaining integral is the integral of the delta function over an interval that includes its non-zero point. According to the second property of the delta function, this integral must equal 1, regardless of the size of the interval as long as it contains the impulse.

$$\int_{t_0-\epsilon}^{t_0+\epsilon} \delta(t - t_0) dt = 1$$

Substituting this back into our expression for I , we get the final result:

$$I = f(t_0) \cdot 1 = f(t_0)$$

This completes the proof of the sifting property.

2 Derivation of Fourier Transform Equations

The Fourier transform is derived by extending the Fourier series from periodic functions to non-periodic functions by taking the limit as the period approaches infinity.

1. Fourier Series for a Periodic Function

A periodic function $f_T(t)$ with period T can be represented by a complex Fourier series:

$$f_T(t) = \sum_{n=-\infty}^{\infty} C_n e^{in\omega_0 t}$$

where $\omega_0 = \frac{2\pi}{T}$ is the fundamental frequency, and the Fourier coefficients are:

$$C_n = \frac{1}{T} \int_{-T/2}^{T/2} f_T(t) e^{-in\omega_0 t} dt$$

2. The Limiting Process: $T \rightarrow \infty$

To extend this to a non-periodic function $f(t)$, we consider it as a single period of an infinitely long periodic function. As $T \rightarrow \infty$:

1. The fundamental frequency spacing $\omega_0 = \frac{2\pi}{T}$ becomes infinitesimally small. We denote this as $d\omega = \frac{2\pi}{T}$.

2. The discrete frequencies $n\omega_0$ become a continuous variable ω .
3. The summation becomes an integral.

Let's rearrange the coefficient equation:

$$TC_n = \int_{-T/2}^{T/2} f_T(t) e^{-in\omega_0 t} dt$$

As $T \rightarrow \infty$, the integral limits extend to $\pm\infty$, and the term TC_n becomes a continuous function of ω , which we define as the **Fourier transform**, $F(\omega)$.

$$F(\omega) = \lim_{T \rightarrow \infty} TC_n = \int_{-\infty}^{\infty} f(t) e^{-i\omega t} dt$$

This is the **forward Fourier transform equation**.

Now, let's substitute C_n back into the Fourier series equation:

$$f_T(t) = \sum_{n=-\infty}^{\infty} \left(\frac{1}{T} \int_{-T/2}^{T/2} f_T(\tau) e^{-in\omega_0 \tau} d\tau \right) e^{in\omega_0 t}$$

Using the identity $\frac{1}{T} = \frac{\omega_0}{2\pi}$, we get:

$$f_T(t) = \frac{1}{2\pi} \sum_{n=-\infty}^{\infty} \left(\int_{-T/2}^{T/2} f_T(\tau) e^{-in\omega_0 \tau} d\tau \right) e^{in\omega_0 t} \omega_0$$

As $T \rightarrow \infty$:

- The term in parentheses becomes the forward Fourier transform, $F(\omega)$.
- The summation $\sum(\dots)\omega_0$ becomes the integral $\int(\dots)d\omega$.

This yields the **inverse Fourier transform equation**:

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) e^{i\omega t} d\omega$$

3 Fourier Transform of a Time-Shifted Impulse

The Fourier transform of a time-shifted impulse function, $\delta(t - t_0)$, is $e^{-j\omega t_0}$. This result shows that shifting a signal in the time domain does not change its frequency content's **magnitude**, but it does introduce a **linear phase shift** proportional to the amount of time shift.

3.1 Derivation

The Fourier transform is defined by the integral:

$$F\{\delta(t - t_0)\} = \int_{-\infty}^{\infty} \delta(t - t_0) e^{-j\omega t} dt$$

This integral can be solved by using the **sifting property** of the Dirac delta function, which states that for any function $f(t)$ that is continuous at t_0 :

$$\int_{-\infty}^{\infty} f(t) \delta(t - t_0) dt = f(t_0)$$

In this case, the function $f(t)$ is $e^{-j\omega t}$. Applying the sifting property, we evaluate this function at the point where the impulse is located, which is $t = t_0$.

$$F\{\delta(t - t_0)\} = e^{-j\omega t} \Big|_{t=t_0} = e^{-j\omega t_0}$$

Therefore, the Fourier transform of a time-shifted impulse is $e^{-j\omega t_0}$.

4 Fourier Transform for an Impulse Train

The Fourier transform of an impulse train, also known as the **Dirac comb**, is another impulse train. This means that a series of equally spaced impulses in the time domain corresponds to a series of equally spaced impulses in the frequency domain.

Equations and Derivations

The impulse train, denoted as $x(t)$, is a periodic signal with a period T . It is represented as an infinite sum of Dirac delta functions, shifted by integer multiples of T .

4.1 Time Domain Representation

The impulse train is defined as:

$$x(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT)$$

where:

- $\delta(t)$ is the **Dirac delta function**, which is an idealized impulse that is zero everywhere except at $t = 0$, where it is infinitely high, and its total area is equal to 1.
- T is the period between impulses.

4.2 Derivation using Fourier Series

Since the impulse train is a periodic signal, we can represent it using the **Fourier series**. The Fourier series of a periodic signal $x(t)$ with period T is given by:

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{j \frac{2\pi k}{T} t}$$

where the Fourier coefficients a_k are calculated as:

$$a_k = \frac{1}{T} \int_{-T/2}^{T/2} x(t) e^{-j \frac{2\pi k}{T} t} dt$$

To find the coefficients for the impulse train, we substitute $x(t) = \delta(t)$ into the integral (we only need to consider one impulse in the interval of integration):

$$a_k = \frac{1}{T} \int_{-T/2}^{T/2} \delta(t) e^{-j \frac{2\pi k}{T} t} dt$$

We can use the **sifting property** of the Dirac delta function, which states that for any function $f(t)$:

$$\int_{-\infty}^{\infty} f(t) \delta(t - t_0) dt = f(t_0)$$

In our case, $f(t) = e^{-j \frac{2\pi k}{T} t}$ and $t_0 = 0$. Applying the sifting property to our integral gives:

$$a_k = \frac{1}{T} (e^{-j \frac{2\pi k}{T} (0)}) = \frac{1}{T} (e^0) = \frac{1}{T}$$

This means all the Fourier series coefficients are the same.

Now, we can find the Fourier transform of the impulse train, $X(\omega)$, by taking the Fourier transform of its Fourier series representation. The Fourier transform of a complex exponential $e^{j\omega_0 t}$ is $2\pi\delta(\omega - \omega_0)$. The Fourier transform of the impulse train is:

$$\begin{aligned} X(\omega) &= \mathcal{F} \left\{ \sum_{k=-\infty}^{\infty} a_k e^{j \frac{2\pi k}{T} t} \right\} \\ X(\omega) &= \sum_{k=-\infty}^{\infty} a_k \mathcal{F} \left\{ e^{j \frac{2\pi k}{T} t} \right\} \end{aligned}$$

Substituting $a_k = \frac{1}{T}$ and the Fourier transform of the exponential:

$$X(\omega) = \sum_{k=-\infty}^{\infty} \frac{1}{T} \left(2\pi \delta\left(\omega - \frac{2\pi k}{T}\right) \right)$$

4.3 Final Result

The final equation for the Fourier transform of an impulse train is:

$$X(\omega) = \frac{2\pi}{T} \sum_{k=-\infty}^{\infty} \delta\left(\omega - \frac{2\pi k}{T}\right)$$

This shows that the Fourier transform is a series of impulses in the frequency domain, with a new period of $\Omega = \frac{2\pi}{T}$. The amplitude of each impulse is $\frac{2\pi}{T}$.

5 The Convolution Theorem

The **convolution theorem** states that the Fourier transform of a convolution of two functions is the pointwise product of their individual Fourier transforms. Mathematically, for two functions $f(t)$ and $g(t)$, the theorem states:

$$\mathcal{F}\{f * g\}(k) = \mathcal{F}\{f\}(k) \cdot \mathcal{F}\{g\}(k)$$

Proof

We start with the definition of the Fourier transform of a convolution:

$$\begin{aligned}\mathcal{F}\{f * g\}(k) &= \int_{-\infty}^{\infty} (f * g)(t) e^{-ikt} dt \\ &\quad \text{(by definition of convolution)} \\ &= \int_{-\infty}^{\infty} \left(\int_{-\infty}^{\infty} f(\tau) g(t - \tau) d\tau \right) e^{-ikt} dt \\ &\quad \text{(change the order of integration)} \\ &= \int_{-\infty}^{\infty} f(\tau) \left(\int_{-\infty}^{\infty} g(t - \tau) e^{-ikt} dt \right) d\tau \\ &\quad \text{(change variables in the inner integral: let } u = t - \tau, \text{ so } t = u + \tau \text{ and } dt = du) \\ &= \int_{-\infty}^{\infty} f(\tau) \left(\int_{-\infty}^{\infty} g(u) e^{-ik(u+\tau)} du \right) d\tau \\ &= \int_{-\infty}^{\infty} f(\tau) \left(\int_{-\infty}^{\infty} g(u) e^{-iku} e^{-ik\tau} du \right) d\tau \\ &\quad \text{(factor out } e^{-ik\tau} \text{ from the inner integral)} \\ &= \int_{-\infty}^{\infty} f(\tau) e^{-ik\tau} \left(\int_{-\infty}^{\infty} g(u) e^{-iku} du \right) d\tau \\ &\quad \text{(recognize the inner integral as } \mathcal{F}\{g\}(k)) \\ &= \int_{-\infty}^{\infty} f(\tau) e^{-ik\tau} \mathcal{F}\{g\}(k) d\tau \\ &\quad \text{(factor out } \mathcal{F}\{g\}(k) \text{ from the outer integral)} \\ &= \mathcal{F}\{g\}(k) \int_{-\infty}^{\infty} f(\tau) e^{-ik\tau} d\tau \\ &\quad \text{(recognize the remaining integral as } \mathcal{F}\{f\}(k)) \\ &= \mathcal{F}\{g\}(k) \cdot \mathcal{F}\{f\}(k)\end{aligned}$$

This completes the proof.

6 DFT

The DFT is the discrete version of the continuous Fourier Transform, which applies to continuous functions.

Derivation

The DFT can be understood by starting with the continuous Fourier Transform and then applying discretization.

1. The Continuous Fourier Transform

The continuous Fourier Transform of a function $x(t)$ is given by:

$$X(f) = \int_{-\infty}^{\infty} x(t)e^{-j2\pi ft} dt$$

This formula calculates the frequency spectrum $X(f)$ of a continuous signal $x(t)$. The term $e^{-j2\pi ft}$ represents a complex sinusoid, and the integral calculates how much of this specific frequency f is present in the signal.

2. From Continuous to Discrete

To derive the DFT, we need to make two key assumptions for a **finite** set of data:

1. **The signal is discrete in time:** We sample the continuous signal $x(t)$ at a fixed interval T_s . This gives us a sequence of N samples, $x_0, x_1, \dots, x_{N-1}$, where $x_n = x(nT_s)$. The total duration of the signal is $T = NT_s$.
2. **The signal is periodic:** We assume the sampled signal repeats with a period of N samples. This allows us to represent the signal as a sum of discrete sinusoids.

Due to the Nyquist-Shannon sampling theorem, the maximum frequency we can resolve is the Nyquist frequency, $f_{\max} = 1/(2T_s)$. The discrete frequency steps in the frequency domain will be $\Delta f = 1/T = 1/(NT_s)$. The discrete frequencies are then $f_k = k\Delta f = k/(NT_s)$, for $k = 0, 1, \dots, N-1$.

3. Substituting into the Formula

Now, we can replace the integral with a summation and the continuous variables with their discrete counterparts.

- The integral $\int$ is replaced by a summation $\sum_{n=0}^{N-1}$.
- The continuous time t is replaced by the discrete time nT_s .
- The continuous frequency f is replaced by the discrete frequency $k/(NT_s)$.
- The differential dt is replaced by the sampling interval T_s .

Substituting these into the continuous Fourier Transform formula:

$$X(f) = \int_{-\infty}^{\infty} x(t)e^{-j2\pi ft} dt$$
$$X(k) \approx \sum_{n=0}^{N-1} x(nT_s)e^{-j2\pi(\frac{k}{NT_s})(nT_s)T_s}$$

The T_s term is a scaling factor. We can omit it in the standard definition of the DFT, as it can be absorbed into the scaling of the output. This gives us the **DFT equation**:

$$X_k = \sum_{n=0}^{N-1} x_n e^{-j2\pi \frac{kn}{N}} \quad \text{for } k = 0, 1, \dots, N-1$$

Here, x_n represents the discrete time-domain samples and X_k represents the discrete frequency-domain coefficients. The term $e^{-j2\pi \frac{kn}{N}}$ is a complex exponential that represents a complex sinusoid at the k -th discrete frequency. The summation calculates the contribution of each time sample to each frequency component.

The **Inverse Discrete Fourier Transform (IDFT)** allows us to reconstruct the original signal from its frequency components:

$$x_n = \frac{1}{N} \sum_{k=0}^{N-1} X_k e^{j2\pi \frac{kn}{N}} \quad \text{for } n = 0, 1, \dots, N-1$$

This formula reconstructs the time-domain signal by summing up the weighted complex sinusoids, where each weight is given by the DFT coefficient X_k . The $1/N$ factor is necessary for proper scaling.

7 FFT

The **Fast Fourier Transform (FFT)** is an efficient algorithm for computing the **Discrete Fourier Transform (DFT)**. In the context of polynomials, it provides a fast method for multiplying two polynomials by converting them from their **coefficient representation** to their **point-value representation**.

The Three-Step Process

1. **Evaluate:** Convert polynomials to their point-value representation using the FFT.
2. **Pointwise Multiplication:** Multiply the corresponding point values.
3. **Interpolate:** Convert the product's point-value representation back to its coefficient representation using the inverse FFT (IFFT).

Example with Two Polynomials

Let's consider two simple polynomials:

$$A(x) = 3x + 1$$

$$B(x) = 2x + 4$$

Their product is $C(x) = A(x)B(x) = (3x + 1)(2x + 4) = 6x^2 + 14x + 4$.

Step 1: Evaluate using FFT

To evaluate at $2n = 4$ points, we use the 4th roots of unity: $1, i, -1, -i$.

For $A(x) = 3x + 1$:

- $A(1) = 3(1) + 1 = 4$
- $A(i) = 3(i) + 1 = 1 + 3i$
- $A(-1) = 3(-1) + 1 = -2$
- $A(-i) = 3(-i) + 1 = 1 - 3i$

Point-value representation of $A(x)$: $(4, 1 + 3i, -2, 1 - 3i)$.

For $B(x) = 2x + 4$:

- $B(1) = 2(1) + 4 = 6$
- $B(i) = 2(i) + 4 = 4 + 2i$
- $B(-1) = 2(-1) + 4 = 2$
- $B(-i) = 2(-i) + 4 = 4 - 2i$

Point-value representation of $B(x)$: $(6, 4 + 2i, 2, 4 - 2i)$.

Step 2: Pointwise Multiplication

We multiply the corresponding point values to find the point-value representation of $C(x)$:

- $C(1) = A(1) \cdot B(1) = 4 \cdot 6 = 24$
- $C(i) = A(i) \cdot B(i) = (1 + 3i)(4 + 2i) = -2 + 14i$
- $C(-1) = A(-1) \cdot B(-1) = (-2)(2) = -4$
- $C(-i) = A(-i) \cdot B(-i) = (1 - 3i)(4 - 2i) = -2 - 14i$

Point-value representation of $C(x)$: $(24, -2 + 14i, -4, -2 - 14i)$.

Step 3: Interpolate using Inverse FFT

The inverse FFT converts the point values back to the coefficients. The coefficients c_k are given by the formula:

$$c_k = \frac{1}{N} \sum_{j=0}^{N-1} y_j (\omega_N^{-k})^j$$

where $N = 4$ and ω_N^{-1} is the inverse root of unity.

The coefficients of $C(x)$ are:

$$c_0 = \frac{1}{4}(24 + (-2 + 14i) + (-4) + (-2 - 14i)) = 4$$

$$c_1 = \frac{1}{4}(24(i^0) + (-2 + 14i)(i^{-1}) + (-4)(i^{-2}) + (-2 - 14i)(i^{-3})) = 14$$

$$c_2 = \frac{1}{4}(24(i^0) + (-2 + 14i)(i^{-2}) + (-4)(i^{-4}) + (-2 - 14i)(i^{-6})) = 6$$

Therefore, the product polynomial is $C(x) = 6x^2 + 14x + 4$, which is the correct result.